LogicLab Programming Logic and Design

Questions only

229 numbered exercises covering the five learning units and task library. Use the same exercise numbers to match questions with the answer pack. Methods use zero-based arrays and inclusive FOR bounds. Written tasks can have other valid solutions.

Open revision material. Attempt each question before consulting the answer pack.

Unit 1 Algorithms and the development life cycle

Exercise 1 Knowledge check 1

What should happen before choosing program statements?
A. Write the final code
B. Clarify inputs, outputs and requirements
C. Deploy the program

Exercise 2 Knowledge check 2

Which activity belongs to requirements analysis?
A. Choose a button colour first
B. Ignore invalid data
C. Identify required inputs and outputs

Exercise 3 Knowledge check 3

Which statement describes an algorithm?
A. A particular graphics card
B. A finite ordered solution procedure
C. Only source code in C#

Exercise 4 Knowledge check 4

When should documentation happen?
A. Throughout development
B. Only after retirement
C. Never during design

Exercise 5 Knowledge check 5

A program runs without crashing. What follows?
A. It must be correct
B. All boundaries are covered
C. Its outputs still need testing

Exercise 6 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. INPUT price
3. INPUT quantity
4. SET total = price + quantity
5. OUTPUT total
6. END
```

Exercise 7 Build or trace a solution

Read a parcel weight and a rate per kilogram. Calculate and display the delivery charge. Assume valid non-negative input. Work independently before comparing your answer.

Exercise 8 Apply the concept

Add a fixed handling charge of R15. Give two test cases.

Unit 1 Pseudocode and flowchart symbols

Exercise 9 Knowledge check 1

Which symbol represents an input operation?
A. Rectangle
B. Diamond
C. Parallelogram

Exercise 10 Knowledge check 2

Which symbol should contain a calculation?
A. Decision diamond
B. Process rectangle
C. Input parallelogram

Exercise 11 Knowledge check 3

What does a flowchart arrow show?
A. Direction of control flow
B. The data type
C. The program title

Exercise 12 Knowledge check 4

Why label decision branches?
A. To sort values
B. To create variables
C. To distinguish condition outcomes

Exercise 13 Knowledge check 5

Is there one universal pseudocode syntax?
A. Only indentation is relevant
B. No, state the chosen conventions
C. Yes, every language uses one

Exercise 14 Find and fix the error

Identify the faulty line, error category and correction.
1. BEGIN
2. INPUT age
3. IF age >= 18 THEN
4. OUTPUT "Adult"
5. ENDWHILE
6. END

Exercise 15 Build or trace a solution

Read a temperature. Display "Cold" below 15 and "Mild or warm" otherwise. Draw the corresponding flowchart. Work independently before comparing your answer.

Exercise 16 Apply the concept

List the correct flowchart shapes for reading a mark, computing an average, testing a pass condition and ending.

Unit 1 Data types, variables and constants

Exercise 17 Knowledge check 1

```
Which is usually the best type for a South African phone number?
A. INTEGER
B. STRING
C. BOOLEAN
```

Exercise 18 Knowledge check 2

```
What does initialisation do?
A. Gives a variable its starting value
B. Ends every loop
C. Changes the identifier type automatically
```

Exercise 19 Knowledge check 3

```
Which type represents TRUE or FALSE?
A. STRING only
B. REAL
C. BOOLEAN
```

Exercise 20 Knowledge check 4

```
What value begins a multiplication accumulator?
A. -1 always
B. 1
C. 0
```

Exercise 21 Knowledge check 5

```
Can an INTEGER age be invalid?
A. Yes, a negative age can violate requirements
B. No, types guarantee all rules
C. Only if its name is age
```

Exercise 22 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. DECLARE total AS REAL
3. INPUT price
4. SET total = total + price
5. OUTPUT total
6. END
```

Exercise 23 Build or trace a solution

Calculate the product of three input numbers using an accumulator. Work independently before comparing your answer.

Exercise 24 Apply the concept

Choose types for student number, active flag, whole-item count and average mark. Justify each.

Unit 1 Expressions and operator precedence

Exercise 25 Knowledge check 1

```
What is 4 + 3 * 2?
A. 14
B. 10
C. 11
```

Exercise 26 Knowledge check 2

```
What is (8 + 4) / 3?
A. 12
B. 9
C. 4
```

Exercise 27 Knowledge check 3

```
What is 17 MOD 5?
A. 5
B. 2
C. 3
```

Exercise 28 Knowledge check 4

```
What is 17 DIV 5?
A. 3
B. 2
C. 3.4
```

Exercise 29 Knowledge check 5

```
Which expression averages a and b?
A. a + b / 2
B. a * b / 2
C. (a + b) / 2
```

Exercise 30 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. INPUT mark1
3. INPUT mark2
4. SET average = mark1 + mark2 / 2
5. OUTPUT average
6. END
```

Exercise 31 Build or trace a solution

Read price and quantity, then apply a 10% discount to the subtotal. Work independently before comparing your answer.

Exercise 32 Apply the concept

Write an expression converting 135 minutes to whole hours and remaining minutes.

Unit 1 Structured design and trace tables

Exercise 33 Knowledge check 1

```
Which structure repeats a block of statements?
A. Sequence
B. Selection
C. Repetition
```

Exercise 34 Knowledge check 2

```
What is a trace table used for?
A. List only method names
B. Record changing state and output
C. Replace every test
```

Exercise 35 Knowledge check 3

```
After x=2 and y=x+5, what is y?
A. 7
B. 2
C. 5
```

Exercise 36 Knowledge check 4

```
Which is a selection structure?
A. A single OUTPUT
B. An assignment alone
C. IF/ELSE
```

Exercise 37 Knowledge check 5

```
Which is a feature of maintainable code?
A. All tasks in one enormous method
B. Clear names and focused responsibilities
C. Maximum duplication
```

Exercise 38 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. SET x = 3
3. SET y = x + 2
4. SET x = y * 2
5. OUTPUT y
6. END
```

Exercise 39 Build or trace a solution

Dry-run the assignments and record the values after each statement. Work independently before comparing your answer.

Exercise 40 Apply the concept

Classify each action: read two marks, choose whether a mark passes, repeat input for ten learners.

Unit 2 Boolean expressions and basic selection

Exercise 41 Knowledge check 1

When is A OR B false?
A. When both are false
B. When either is true
C. When both are true

Exercise 42 Knowledge check 2

When is A AND B true?
A. Only when both are true
B. When either is true
C. When both are false

Exercise 43 Knowledge check 3

What is NOT FALSE?
A. FALSE
B. Undefined
C. TRUE

Exercise 44 Knowledge check 4

Which condition accepts 0 through 100?
A. mark > 0 AND mark < 100
B. mark >= 0 AND mark <= 100
C. mark >= 0 OR mark <= 100

Exercise 45 Knowledge check 5

Which values best test a pass boundary of 50?
A. 49, 50 and 51
B. 1, 2 and 3
C. Only 100

Exercise 46 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. INPUT mark
3. IF mark &gt; 50 THEN
4. OUTPUT "Pass"
5. ELSE
6. OUTPUT "Not yet passed"
7. ENDIF
8. END
```

Exercise 47 Build or trace a solution

Read a mark and display whether it is between 0 and 100 inclusive. Work independently before comparing your answer.

Exercise 48 Apply the concept

Write an invalid-range condition using OR, and another using NOT.

Unit 2 Precedence and nested decisions

Exercise 49 Knowledge check 1

What is TRUE OR FALSE AND FALSE?
A. FALSE
B. TRUE
C. Invalid

Exercise 50 Knowledge check 2

Which logical operator has highest precedence here?
A. OR
B. AND
C. NOT

Exercise 51 Knowledge check 3

In nested IF, when is the inner branch considered?
A. Only at program end
B. Only after execution enters its outer branch
C. Before every outer condition

Exercise 52 Knowledge check 4

Why test a 75 distinction before a 50 pass band?
A. The more specific higher band should be checked first
B. It changes the input type
C. It removes the need for ELSE

Exercise 53 Knowledge check 5

What does CASE typically compare?
A. Only two variables without values
B. Every array position automatically
C. One expression against listed values

Exercise 54 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. INPUT age
3. INPUT hasID
4. IF age &gt;= 18 OR hasID = TRUE THEN
5. OUTPUT "Entry allowed"
6. ENDIF
7. END
```

Exercise 55 Build or trace a solution

Display Distinction for marks of at least 75, Pass for marks of at least 50, otherwise Revise. Work independently before comparing your answer.

Exercise 56 Apply the concept

Rewrite the band selection using a nested IF inside ELSE.

Unit 2 Counted loops and accumulators

Exercise 57 Knowledge check 1

How many times does FOR i = 1 TO 4 execute with STEP 1?
A. 3
B. 4
C. 5

Exercise 58 Knowledge check 2

What is a counter normally increased by when counting occurrences?
A. 0
B. 1
C. The total input value

Exercise 59 Knowledge check 3

When should a final grand total usually be displayed?
A. After accumulation finishes
B. Before initialisation
C. Only inside the first iteration

Exercise 60 Knowledge check 4

How many iterations does FOR i=2 TO 6 STEP 2 have?
A. 2
B. 6
C. 3

Exercise 61 Knowledge check 5

How many iterations does FOR i=5 TO 1 STEP -1 have?
A. 0
B. 5
C. 4

Exercise 62 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. SET total = 0
3. FOR i = 1 TO 3
4. INPUT value
5. SET total = value
6. ENDFOR
7. OUTPUT total
8. END
```

Exercise 63 Build or trace a solution

Read four marks and count how many are at least 50. Work independently before comparing your answer.

Exercise 64 Apply the concept

Calculate the average of four marks. Include initialisation and final output.

Unit 2 Conditional loops and priming input

Exercise 65 Knowledge check 1

```
Which loop always executes its body at least once?
A. WHILE
B. REPEAT/UNTIL
C. Every FOR loop
```

Exercise 66 Knowledge check 2

```
Can a WHILE body run zero times?
A. Yes
B. No
C. Only with strings
```

Exercise 67 Knowledge check 3

```
What should happen to a sentinel value?
A. It must be added to every total
B. It becomes the average
C. It should normally be excluded from data totals
```

Exercise 68 Knowledge check 4

```
When does REPEAT/UNTIL stop?
A. Before the first body execution
B. When its condition is true
C. While its condition is false before entry
```

Exercise 69 Knowledge check 5

```
Why guard average calculation when count is zero?
A. To avoid division by zero
B. To change count into an array
C. To force one more valid value
```

Exercise 70 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. INPUT value
3. WHILE value &lt;&gt; -1 DO
4. OUTPUT value
5. SET value = value
6. ENDWHILE
7. END
```

Exercise 71 Build or trace a solution

Read values until -1. Display their average, or No data when the first input is -1. Work independently before comparing your answer.

Exercise 72 Apply the concept

Validate a non-negative age with a post-test loop.

Unit 2 Nested loops

Exercise 73 Knowledge check 1

How many pairs are output by the example?
A. 2
B. 3
C. 6

Exercise 74 Knowledge check 2

A 4 by 3 nested traversal visits how many cells?
A. 7
B. 4
C. 12

Exercise 75 Knowledge check 3

Where does a per-row total reset?
A. At every inner comparison after adding
B. At the start of each outer iteration
C. Only once after both loops

Exercise 76 Knowledge check 4

Why use distinct nested-loop counters?
A. So the inner loop does not overwrite the outer state
B. To prevent all arithmetic
C. Because counters must be text

Exercise 77 Knowledge check 5

The inner loop runs 5 times for each of 2 outer iterations. Body count?
A. 5
B. 2
C. 10

Exercise 78 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. FOR row = 1 TO 2
3. FOR row = 1 TO 3
4. OUTPUT row
5. ENDFOR
6. ENDFOR
7. END
```

Exercise 79 Build or trace a solution

Output the products for rows 1 to 3 and columns 1 to 2. Work independently before comparing your answer.

Exercise 80 Apply the concept

Where should a row total and a grand total be initialised?

Unit 3 Arrays and element access

Exercise 81 Knowledge check 1

What is the last valid index in a zero-based array of length 5?
A. 5
B. 4
C. 6

Exercise 82 Knowledge check 2

What is the first index in this platform?
A. -1
B. 0
C. 1

Exercise 83 Knowledge check 3

For n values, which bound is correct?
A. 0 TO n - 1
B. 0 TO n
C. 1 TO n + 1

Exercise 84 Knowledge check 4

What accesses row r, column c?
A. values[r + c] always
B. values[n]
C. values[r][c]

Exercise 85 Knowledge check 5

How does capacity differ from meaningful count?
A. Capacity is a variable name only
B. Capacity can exceed the number of used values
C. They must always be equal

Exercise 86 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. FOR i = 0 TO 5
3. OUTPUT values[i]
4. ENDFOR
5. END
```

Exercise 87 Build or trace a solution

Total an existing zero-based array containing five values. Work independently before comparing your answer.

Exercise 88 Apply the concept

Write a traversal for a 2 by 3 zero-based array.

Unit 3 Linear search

Exercise 89 Knowledge check 1

```
Why initialise foundAt to -1?
A. It is a valid first index
B. It distinguishes not found from index 0
C. It sorts the array
```

Exercise 90 Knowledge check 2

```
Does linear search require sorted data?
A. No
B. Yes, always
C. Only in ascending order
```

Exercise 91 Knowledge check 3

```
Worst-case linear search comparisons for n items?
A. 1 always
B. 0
C. n
```

Exercise 92 Knowledge check 4

```
How do you find every occurrence?
A. Only compare index zero
B. Continue through all valid positions
C. Stop at the first match
```

Exercise 93 Knowledge check 5

```
What should happen before using foundAt as an index?
A. Check that a match was found
B. Replace it with zero
C. Add one without checking
```

Exercise 94 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. SET foundAt = 0
3. FOR i = 0 TO 3
4. IF values[i] = target THEN
5. SET foundAt = i
6. ENDIF
7. ENDFOR
8. OUTPUT foundAt
9. END
```

Exercise 95 Build or trace a solution

Find the first position of target in a four-value array, stopping after a match. Work independently before comparing your answer.

Exercise 96 Apply the concept

How would you find every matching position rather than only the first?

Unit 3 Parallel arrays

Exercise 97 Knowledge check 1

Which name belongs to scores[2]?
A. names[0]
B. names[2]
C. The largest name

Exercise 98 Knowledge check 2

What links parallel arrays?
A. Identical variable names
B. The largest array value
C. A shared record index

Exercise 99 Knowledge check 3

What must happen when a name is swapped?
A. Nothing else may change
B. Its related attributes must move too
C. All marks must be cleared

Exercise 100 Knowledge check 4

Which design can reduce parallel-array alignment errors?
A. Objects or records holding related attributes
B. More unrelated global variables
C. Sorting just one array

Exercise 101 Knowledge check 5

names[1] belongs with which score?
A. scores[0] always
B. The highest score
C. scores[1]

Exercise 102 Find and fix the error

Identify the faulty line, error category and correction.

```
1. BEGIN
2. SET i = 2
3. OUTPUT names[i], scores[0]
4. END
```

Exercise 103 Build or trace a solution

Search three learner names and output the matching mark. Work independently before comparing your answer.

Exercise 104 Apply the concept

Explain how to swap two parallel records at positions j and j+1.

Unit 3 Bubble sort

Exercise 105 Knowledge check 1

```
Why use a temporary variable during a swap?
A. To preserve a value before overwriting it
B. To increase the array length
C. To avoid comparing values
```

Exercise 106 Knowledge check 2

```
After a full ascending left-to-right bubble pass, where is the largest value?
A. Removed from the array
B. At the end of the unsorted part
C. Always at index zero
```

Exercise 107 Knowledge check 3

```
Maximum basic passes for n values?
A. n - 1
B. n + 1
C. 1 for every input
```

Exercise 108 Knowledge check 4

```
Which adjacent comparison causes an ascending swap?
A. values[j] &lt; values[j + 1]
B. values[j] = values[j + 1]
C. values[j] &gt; values[j + 1]
```

Exercise 109 Knowledge check 5

```
What can a swapped flag improve?
A. Increase array capacity
B. Stop after a pass with no swaps
C. Guarantee zero comparisons
```

Exercise 110 Find and fix the error

```
Identify the faulty line, error category and correction.
1. BEGIN
2. SET temp = values[j]
3. SET values[j] = values[j + 1]
4. SET values[j + 1] = values[j]
5. END
```

Exercise 111 Build or trace a solution

```
Perform one left-to-right ascending pass through four existing values. Work independently
 before comparing your answer.
```

Exercise 112 Apply the concept

Give a full bubble sort for a zero-based array with n values.

Unit 4 Decomposition and hierarchy charts

Exercise 113 Knowledge check 1

What does a hierarchy chart primarily show?
A. Every loop iteration
B. Module organisation and calls
C. The current values of all variables

Exercise 114 Knowledge check 2

Which is good decomposition?
A. Separate focused responsibilities
B. Put every operation in one module
C. Use one method for unrelated tasks

Exercise 115 Knowledge check 3

What describes cohesion?
A. How many fonts a page uses
B. The total variable count only
C. How closely a module’s tasks belong together

Exercise 116 Knowledge check 4

Does a hierarchy chart show every Boolean path?
A. Only if methods return integers
B. No, it shows module organisation
C. Yes, that is its main purpose

Exercise 117 Knowledge check 5

What is coupling?
A. Dependencies between modules
B. The number of array elements
C. The multiplication operator

Exercise 118 Find and fix the error

Identify the faulty line, error category and correction.

```
1. METHOD showResult()
2. OUTPUT "Result ready"
3. ENDMETHOD
4. BEGIN
5. CALL showResults()
6. END
```

Exercise 119 Build or trace a solution

Create separate modules for greeting and displaying a learning reminder. Work independently before comparing your answer.

Exercise 120 Apply the concept

Draw a hierarchy where Main calls calculate and calculate calls validate. Describe what the links mean.

Unit 4 Parameters, arguments and scope

Exercise 121 Knowledge check 1

```
In CALL displayName("Lebo"), what is "Lebo"?
A. A parameter declaration
B. An argument
C. A method header
```

Exercise 122 Knowledge check 2

```
What is a parameter?
A. Only a literal at the call site
B. The final program output
C. A placeholder in a method definition
```

Exercise 123 Knowledge check 3

```
What is local scope?
A. The whole computer network
B. The region where a local identifier is accessible
C. An automatic global variable
```

Exercise 124 Knowledge check 4

```
For pass-by-value INTEGER x, local reassignment changes what?
A. The local parameter copy
B. The caller variable automatically
C. Every integer in the program
```

Exercise 125 Knowledge check 5

```
What should match at a call?
A. Only parameter spelling
B. Only return-value colour
C. Argument count, order and compatible types
```

Exercise 126 Find and fix the error

```
Identify the faulty line, error category and correction.
1. METHOD displayPair(first, second)
2. OUTPUT first, second
3. ENDMETHOD
4. BEGIN
5. CALL displayPair("Lebo")
6. END
```

Exercise 127 Build or trace a solution

Write a method that displays a supplied learner name and mark. Work independently before comparing your answer.

Exercise 128 Apply the concept

Explain what happens with pass-by-value when a method reassigns an INTEGER parameter.

Unit 4 Functions and return values

Exercise 129 Knowledge check 1

Which statement gives a function result back to its caller?
A. OUTPUT result
B. RETURN result
C. INPUT result

Exercise 130 Knowledge check 2

Does OUTPUT necessarily return a usable function value?
A. Only for positive numbers
B. No
C. Yes, always

Exercise 131 Knowledge check 3

What does RETURN do to its method invocation?
A. Supplies a result and exits that invocation
B. Always restarts it
C. Displays every local variable

Exercise 132 Knowledge check 4

A function promises a result. Which paths need one?
A. Only the first branch
B. Only the ELSE branch
C. Every normal finishing path

Exercise 133 Knowledge check 5

Where can a returned result be used?
A. Only in a flowchart title
B. In an assignment or expression
C. Only in a comment

Exercise 134 Find and fix the error

Identify the faulty line, error category and correction.
1. FUNCTION add(a, b) RETURNS REAL
2. SET result = a + b
3. OUTPUT result
4. ENDFUNCTION

Exercise 135 Build or trace a solution

Write a method returning the larger of two values. Include an early return in each branch. Work independently before comparing your answer.

Exercise 136 Apply the concept

Write a method that returns whether a number is even.

Unit 4 Passing arrays to methods

Exercise 137 Knowledge check 1

Why might a method need count as well as the array?
A. Count may differ from capacity
B. Arrays cannot contain numbers
C. It removes the need for bounds

Exercise 138 Knowledge check 2

Which count bound should an array method require?
A. 0 <= count <= capacity
B. count > capacity
C. count must be -1

Exercise 139 Knowledge check 3

Should array mutation assumptions be documented?
A. No
B. Only for sorted arrays
C. Yes

Exercise 140 Knowledge check 4

When a C# array reference is passed by value, can elements be updated?
A. Only if the array is empty
B. Yes, the reference copy can refer to the same array
C. No, all elements are copied automatically

Exercise 141 Knowledge check 5

For count=3, which indices are processed?
A. 0, 1, 2
B. 1, 2, 3, 4
C. 0, 1, 2, 3

Exercise 142 Find and fix the error

Identify the faulty line, error category and correction.

```
1. METHOD displayValues(values, count)
2. FOR i = 0 TO count
3. OUTPUT values[i]
4. ENDFOR
5. ENDMETHOD
```

Exercise 143 Build or trace a solution

Pass an array and meaningful count to a method that returns their total. Work independently before comparing your answer.

Exercise 144 Apply the concept

What preconditions should an array-processing method document?

Unit 4 Overloading and method contracts

Exercise 145 Knowledge check 1

```
Which change can create a valid overload?
A. Change only the return type
B. Change the parameter types
C. Change only parameter names
```

Exercise 146 Knowledge check 2

```
Do parameter names alone distinguish overloads?
A. Yes
B. Only when uppercase
C. No
```

Exercise 147 Knowledge check 3

```
Can return type alone distinguish a C# overload?
A. Only when REAL
B. No
C. Yes
```

Exercise 148 Knowledge check 4

```
Which pair is distinct?
A. sum(a,b) and sum(a,b,c)
B. sum(INTEGER a) and sum(INTEGER b)
C. Two methods with identical parameter types
```

Exercise 149 Knowledge check 5

```
What should overloaded operations usually share?
A. Completely unrelated behaviour
B. No understandable purpose
C. A closely related responsibility
```

Exercise 150 Find and fix the error

```
Identify the faulty line, error category and correction.
1. FUNCTION calculate(INTEGER a) RETURNS INTEGER
2. RETURN a
3. ENDFUNCTION
4. FUNCTION calculate(INTEGER b) RETURNS REAL
5. RETURN b
6. ENDFUNCTION
```

Exercise 151 Build or trace a solution

Compare two signatures and identify the selected overload for two calls. Work independently before comparing your answer.

Exercise 152 Apply the concept

Design two valid overloads for a total method using different parameter counts.

Unit 5 Classes, objects and constructors

Exercise 153 Knowledge check 1

```
Which is an object rather than a class?
A. The Learner type definition
B. One particular Learner instance
C. The STRING type
```

Exercise 154 Knowledge check 2

```
What does a constructor normally establish?
A. The user’s browser type
B. An initial valid object state
C. Every object’s final report
```

Exercise 155 Knowledge check 3

```
Can two objects share equal values yet be distinct?
A. Yes
B. No
C. Only if they are different classes
```

Exercise 156 Knowledge check 4

```
What describes an attribute?
A. Only an external file
B. A loop terminator
C. Object state
```

Exercise 157 Knowledge check 5

```
What describes a method?
A. A number type
B. Behaviour associated with a class or module
C. Only the object’s colour
```

Exercise 158 Find and fix the error

```
Identify the faulty line, error category and correction.
1. CLASS Learner
2. PRIVATE name AS STRING
3. METHOD initialise(newName)
4. SET name = name
5. ENDMETHOD
6. ENDCLASS
```

Exercise 159 Build or trace a solution

Model two Book objects created from the same class. Describe their independent state.
Work independently before comparing your answer.

Exercise 160 Apply the concept

Describe a constructor for Book that requires a title and starts with borrowed = FALSE.

Unit 5 Encapsulation, inheritance and polymorphism

Exercise 161 Knowledge check 1

A Course contains Learner objects. Which relationship is suggested?
A. Course is a Learner
B. Course has learners
C. Learner is a number

Exercise 162 Knowledge check 2

What is encapsulation intended to protect?
A. Valid state through controlled access
B. Only variable spelling
C. Every class from having methods

Exercise 163 Knowledge check 3

Which phrase indicates inheritance?
A. Has-a in every case
B. Prints-a
C. Is-a

Exercise 164 Knowledge check 4

What is overriding?
A. Creating an unrelated class variable
B. Specialising an inherited operation with the same contract
C. Changing only a parameter name

Exercise 165 Knowledge check 5

What enables runtime polymorphic behaviour?
A. A common abstraction with specialised implementations
B. Sorting the object names
C. Making every attribute public

Exercise 166 Find and fix the error

Identify the faulty line, error category and correction.
1. CLASS Account
2. PUBLIC balance AS REAL
3. METHOD deposit(amount)
4. IF amount > 0 THEN
5. SET balance = balance + amount
6. ENDIF
7. ENDMETHOD
8. ENDCLASS

Exercise 167 Build or trace a solution

Compare inheritance and containment in a school model. Work independently before comparing your answer.

Exercise 168 Apply the concept

Explain polymorphism with Shape, Circle and Rectangle without confusing it with overloading.

Unit 5 Class diagrams and class organisation

Exercise 169 Knowledge check 1

In a basic UML class diagram, what does - indicate?
A. Public visibility
B. Private visibility
C. An arithmetic error

Exercise 170 Knowledge check 2

What does + indicate in a basic UML class diagram?
A. Private visibility
B. An addition instruction to execute
C. Public visibility

Exercise 171 Knowledge check 3

What does 0..* indicate?
A. A negative array index
B. Zero or more related objects
C. Exactly one object

Exercise 172 Knowledge check 4

Where does an inheritance triangle point?
A. The general superclass
B. The specialised subclass
C. Both directions simultaneously

Exercise 173 Knowledge check 5

Which diagram fits a class’s attributes and operations?
A. A procedural call hierarchy only
B. A loop trace table only
C. Class diagram

Exercise 174 Find and fix the error

Identify the faulty line, error category and correction.
1. Class diagram: SavingsAccount inherits Account
2. Inheritance triangle points to SavingsAccount

Exercise 175 Build or trace a solution

Describe a Library associated with zero or more Books and its public borrow operation.
Work independently before comparing your answer.

Exercise 176 Apply the concept

Describe the direction of the inheritance triangle between SavingsAccount and Account.

Task library Methods

Exercise 177 Return the larger value

Write Maximum(first, second) to return the larger of two integers. Return either when equal. Call it with (17, 9) and (8, 8).

Exercise 178 A method that counts

Write a method DisplayNumbers() that displays 1 to 5 using WHILE. Call it from the main program.

Exercise 179 Rectangle area

Write RectangleArea(length, width) to return an area. Assume non-negative dimensions. Show calls for (6, 4) and (0, 4).

Exercise 180 Input and average five marks

Write a method that inputs exactly five marks using FOR and returns their average. Call it and display the returned value. Assume valid marks.

Exercise 181 RETURN or OUTPUT?

Explain RETURN versus OUTPUT. Write Square(number), then display Square(6).

Exercise 182 Average through parameters

Write CalculateAverage(a, b, c) to return the average of three REAL marks. Test 60, 75 and 90.

Exercise 183 A Boolean returning method

Write HasPassed(mark) returning TRUE for marks of at least 50. Use it in an IF for 48. Assume valid marks.

Exercise 184 Return a factorial

Write Factorial(number) with a FOR loop. Assume a non-negative integer small enough to avoid overflow. Explain the case number = 0.

Task library Overloading

Exercise 185 Overload by parameter count

Create two Add methods: one accepts two integers, the other three. Show calls and expected outputs.

Exercise 186 Overload by parameter type

Write Multiply for two INTEGER arguments and for two REAL arguments. Explain how the caller selects a method.

Exercise 187 One name, two shapes

Write CalculateArea for a square and a rectangle using overloaded methods. Explain the distinguishing parameter lists.

Exercise 188 Trace nested calls

Calculate(a, b) returns a + b. Calculate(a, b, c) returns a * b * c. Trace Calculate(2, 3), Calculate(2, 3, 4) and Calculate(Calculate(1, 2), 5). Explain each selection.

Exercise 189 Repair an invalid overload

GetPrice(price AS REAL) RETURNS REAL and GetPrice(price AS REAL) RETURNS INTEGER have identical parameter lists. Explain the problem. Design a discount version with a valid overload.

Task library Loops

Exercise 190 Count students who passed

Input a positive student count and each mark. Count marks at least 50. Write a FOR solution, then rewrite it using WHILE. Assume valid input.

Exercise 191 Count even numbers

Input a positive number count and each integer. Count the even integers. Write a FOR solution, then rewrite it using WHILE. Assume valid input.

Exercise 192 Count salaries above a threshold

Input a positive employee count and each salary. Count salaries strictly above R15 000. Write a FOR solution, then rewrite it using WHILE. Assume valid input.

Exercise 193 Calculate a class average

Input a student count n. If n is positive, read n marks and display their average. Handle n = 0. Write FOR and WHILE versions.

Exercise 194 Positive, negative or zero?

Read n integers for a positive n. Count positive and negative values separately. Zero belongs to neither count. Give FOR and WHILE versions.

Task library Parallel arrays

Exercise 195 Capture names and marks

Use parallel arrays to input names and marks for five students. Then display each name with its corresponding mark. Use zero-based indices.

Exercise 196 Filter employee salaries

Input names and salaries of four employees into parallel arrays. Display only employees earning more than R20 000.

Exercise 197 Show all passing students

Given names [Lebo, Thabo, Naledi, Peter, Maria] and marks [65, 48, 72, 55, 81], display all names and marks with a mark of at least 50. Give a FOR solution and explain its WHILE conversion.

Exercise 198 Search by student name

Use names ["Lebo", "Thabo", "Naledi", "Peter", "Maria"] and marks [65, 48, 72, 55, 81]. Ask for a name. Display its paired value when found, otherwise display Not found. Use a WHILE loop; assume unique, case-sensitive names.

Exercise 199 Find a product price

Use names ["Keyboard", "Mouse", "Monitor", "Printer", "Laptop"] and prices [250, 150, 2500, 1800, 9500]. Ask for a name. Display its paired value when found, otherwise display Not found. Use a WHILE loop; assume unique, case-sensitive names.

Exercise 200 Find the top mark

Use the five student names and marks [65, 48, 72, 55, 81]. Find the highest mark and its matching name. Keep the first student if there is a tie.

Exercise 201 A complete results report

Use names [John, Mary, David, Sarah, Peter] and marks [45, 78, 66, 82, 39]. Display every pair, count passes and failures (pass >= 50), and show the top student. Assume a non-empty class.

Task library Revision

Exercise 202 Build a salary flowchart

For a practice sales report, input a basic salary, sales total and commission percentage. Calculate salary = basic + sales * percentage / 100. Write pseudocode and create a flowchart. Test basic 5000, sales 20000 and rate 5.

Exercise 203 Stop on a sentinel

Read numbers while they are greater than zero. Display three times each positive input. Stop when zero or a negative number is entered, without displaying its triple.

Exercise 204 Explain Boolean decisions

A fictional club offers a discount to members aged 65 or older. Write the Boolean condition. Explain AND, OR and NOT using this scenario. State the outcome for a member aged 64 and a non-member aged 70.

Exercise 205 Return a practice contribution

For a fictional exercise only, a company uses a contribution rate of 3%. Write Contribution(salary, rate) returning salary * rate / 100. Call it for salary 10000 and rate 3. This is a programming example, not a real payroll rule.

Exercise 206 Plan before coding

In a fictional leave exercise, staff earn 1.5 days per month worked. Assume monthsRemaining is already known. Identify input, processing and output, then give two test cases.

Task library OOP

Exercise 207 A class that prints numbers

Design NumberPrinter with DisplayNumbers() using FOR to display 1 to 10. Show an object creation and method call.

Exercise 208 Even numbers through an object

Design EvenPrinter with a WHILE method displaying even numbers from 2 to 20. Create an object and call it.

Exercise 209 A class with an array

Design MarkBook with a private marks array containing 60, 75 and 90. A public method must display all marks with FOR. Explain encapsulation.

Exercise 210 Return a total from a class

Design NumberStore with values [4, 6, 10]. Use a WHILE method Total() returning the sum. Explain the counter and accumulator.

Exercise 211 Model a person and student

Design Person with a protected address and Student inheriting from Person with number, firstName and surname. Explain protected versus private. Sketch a class relationship.

Task library Arrays

Exercise 212 Visit every array element

Declare an INTEGER array of five elements. Input each element, then display every value with its zero-based index.

Exercise 213 Array total and average

Given values [12, 8, 20, 10], compute their total and average using a loop. Explain the accumulator.

Exercise 214 Maximum in a single array

Find the maximum of [-8, -3, -12, -5]. Initialise from an array element so the solution works for negative values.

Exercise 215 Find every matching index

Search [4, 7, 4, 9, 4] for an input target. Display every matching index and the total match count.

Exercise 216 Reverse an array in place

Reverse [2, 4, 6, 8, 10] in place using a left and right index. Use a temporary variable when swapping.

Exercise 217 Traverse a two-dimensional array

Input a 2 by 3 INTEGER matrix. Use nested loops to calculate and display the sum of each row. State valid row and column indices.

Task library IF / ELSE

Exercise 218 IF ELSE pass decision

Input a valid mark from 0 to 100. Display Pass if it is at least 50, otherwise display Revise.

Exercise 219 Odd or even

Input an integer. Use IF ELSE and MOD to report whether it is even or odd.

Exercise 220 Validate a mark range

Read a mark. Display Valid only when the mark lies between 0 and 100 inclusive, otherwise display Invalid. Use AND.

Exercise 221 An ELSEIF grade ladder

Assume a valid mark. Display Distinction for 75 or above, Pass for 50 to 74, and Revise below 50. Use IF ELSEIF ELSE.

Exercise 222 Nested eligibility decisions

A fictional activity requires age at least 16 and guardian approval. Use nested IF statements to display Too young, Approval required or Eligible. Assume age is valid and approval is Boolean.

Exercise 223 Calculate a conditional discount

Read a non-negative total and a Boolean membership value. Members spending at least 500 receive 10% off. Everyone else pays the original total. Display the amount due.

Task library SWITCH / CASE

Exercise 224 SWITCH menu

Input menu option 1, 2 or 3. Display Add, View or Exit. Other values must display Invalid option. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.

Exercise 225 Day name lookup

Input a number from 1 to 3 to select Monday, Tuesday or Wednesday. Use DEFAULT for unsupported values. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.

Exercise 226 Text values in SWITCH

Input size S, M or L. Display Small, Medium or Large. Use case-sensitive exact matches and handle unsupported sizes. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.

Exercise 227 A small calculator

Read two numbers and an operator + or *. Use SWITCH to display their sum or product. Reject other operators. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.

Exercise 228 Select a delivery charge

Input zone 1, 2 or 3. Their delivery charges are R30, R50 and R80 respectively. Display the charge or Invalid zone. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.

Exercise 229 Trace a SWITCH decision

Create a SWITCH for option 1 = Start and 2 = Help. Other values display Try again. Trace inputs 2 and 9 and explain why exactly one branch runs. In this pseudocode dialect branches do not fall through and BREAK is unnecessary.